# DIVYANSHU GOYAL

Multimodal Foundation Models | RL Post-Training | Agentic Evaluation

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## SUMMARY

AI researcher and engineering leader with **8+ years of experience** specializing in **agentic AI and Multimodal LLMs**. I bridge the gap between cutting-edge research and enterprise-scale products, specifically in domains where quality is dictated by nuanced human aesthetics and marketing effectiveness. During my tenure at Adobe, I led the technical strategy to scale **GenStudio for Performance Marketing from 0 to 1**, driving **multi-million-dollar adoption** across the Fortune 500. Currently, I am **tech lead for a 7-person team** building specialized VLMs, judge models, and retouchers. Combining product execution with rigorous research, my work includes co-leading the ATune photo retoucher (**outperforming GPT-6 Astra** in human preference), **6 patents**, and publications at CVPR, ICML, and AAAI workshops.

## EXPERIENCE

**Applied Scientist** – Adobe Inc., San Jose, CA June 2021 – Present

*FixIt: Multimodal Models for Marketing & Visual Quality (Tech Lead, Team of 7)*

- Spearheaded technical vision and execution across the **full ML stack** for a **7-person team**, encompassing distributed data pipelines, RL alignment, and low-latency inference.
- Architected a **sub-second** JEV-style judge model integrated directly into the **Firefly AI Agent evaluation harness**, gating agent outputs to reduce user revision iterations by **40%**.
- Directed the deployment of **three specialized model streams**: domain-adapted VLMs, dense reward judges, and an **unsupervised photo retoucher** trained entirely without human-edited pairs.

*Adobe Marketing VLM (Gemma-3-12B, SFT & GRPO Alignment)*

- Fine-tuned **Gemma-3-12B** via distributed SFT followed by multi-round GRPO on 8× H100s, leveraging LoRA at rank 256 (1.19B trainable parameters) across 7 custom reward functions.
- Diagnosed **policy collapse** in marketing quality: the policy predicted the middle rating for 84% of images vs. 56% for humans, because a central guess is never more than 2 points from any label. Mitigated it with a **cubed distance penalty** and **class-weighted advantages** for rare ratings, cutting middle-rating predictions to **64%**.
- Built a **fault-tolerant** annotation pipeline that labeled ∼**10M images** with a teacher VLM to create the SFT dataset.
- Built a **Photoshop agent** powered by the marketing VLM that edits photos toward **marketing taste**, turning the model's judgment of what makes an image effective into concrete Photoshop edits.

*JEV-Style Marketing Judge Models*

- Trained **Qwen3.5-4B** LoRA reward models on a **1.1M soft-labeled dataset** curated from 14 human-rated sources, raising KonIQ correlation to **0.663 SRCC** (vs. 0.421 for the reference model).
- Curated a **67K-item held-out benchmark**, leak-checked against all training images via SSCD copy detection, and gated **10 model releases** on a pre-registered non-inferiority test.
- Conceived and built a **live demo suite** that turns the judge from a scorer into a **creative engine**: it auto-crops images (**0.82 pairwise accuracy on GAIC** vs. 0.71 for the score-based baseline), retouches photos, scores marketing and custom attributes, re-ranks Adobe Stock for marketing use, and flags low-quality assets. Presented it to **leadership and external partners**, opening new customer engagements.
- Caught a run whose vision encoder collapsed within 200 steps, leaving every checkpoint blind to the image, and added an automated **vision-health check** to the training pipeline.

*ATune: Unsupervised Photo Retouching (Co-Lead with Adobe Research)*

- Co-led project design with Adobe Research as co-first author, owning the synthetic data pipeline, training recipe, and human evaluation framework, and **mentored an intern** on the project.
- Trained a **54M-parameter model** that performs non-destructive photo adjustments via off-the-shelf editors (RawTherapee/darktable), learned from **10M unedited photographs with zero expert targets**.
- In a 157-rater human study (3,524 judgments), **beat GPT-6 Astra** (54.3% of decisive votes) and **matched Claude Opus 5** (50.5%); preferred over GPT-6 Astra by at least 8 of 9 published quality judges across FiveK, PPR10K, and SPAQ.
- Ran data-scaling studies from 1M to 10M images: each larger corpus wins or ties, and no judge on any test set prefers a smaller one.

*Adobe GenStudio for Performance Marketing (Tech Lead, Team of 20+)*

- Took GenStudio from **prototype to a multi-million-dollar enterprise product** (launched Oct 2024), leading the production API layer that serves its LLM-powered features to **Fortune 500** customers.
- Co-invented automated brand-guideline extraction and compliance scoring (US20250117829), the foundational engine for **Adobe Brand Intelligence**.
- Developed parameter-efficient fine-tuning pipelines on a Fortune 500 customer's brand data, directly closing a **custom-model enterprise deal**.

*Evaluation Research*

- **DistortBench** (First Author; CVPR/ICML 2026 Workshops): Established a 13.5K-question visual distortion benchmark across 27 types and 5 severity levels; the best of 18 VLMs (61.9%) **trails a human majority vote** (65.7%).
- **JudgeBreak** (ICML 2026 Workshop): Showed that linear probing on hidden representations of Qwen3-32B outperforms LLM reward models and judges by **8 points** (78.1% vs. 70.1%) on subjective evaluations.

**Machine Learning Engineer** – Adobe Inc., Bangalore, India July 2016 – August 2019

- Joined Adobe's core ML serving platform team at inception and helped plan the end-to-end serving infrastructure for internal workloads, including field-aware factorization machines.
- Scaled production inference throughput to **3,500 QPS per node at 0.3–0.6 ms p99 latency** (Adobe Tech Blog), implementing a garbage-collection-free asynchronous event pipeline on LMAX Disruptor.
- Built the model lifecycle platform for versioned deployment, monitoring, and staged rollout.

## PUBLICATIONS

**ATune: Unsupervised Photo Retouching Through Image-Editor Controls**
**D. Goyal***, G. Ruiz*, A. Meisami | Under review | *equal contribution

**DistortBench: Benchmarking Vision Language Models on Image Distortion Identification**
**D. Goyal**, A. Eppa, V. Bannihatti Kumar | CVPR 2026 and ICML 2026 Workshops | arXiv:2604.19966

**JudgeBreak: Is Taste Hidden in the Layers? Stress-Testing Reward Models and LLM Judges with Representation-Based Subjective Evaluation**
V. Bannihatti Kumar, A. Narang, M. Anand, A. Eppa, **D. Goyal** | ICML 2026 Workshop | OpenReview

**Curiosity-Driven LLM-as-a-judge for Personalized Creative Judgment**
V. Bannihatti Kumar, **D. Goyal**, A. Eppa, N. Bhandari | AAAI 2026 Workshop, Oral | arXiv:2510.05135

## PATENTS

| Title | Number | Year |
|---|---|---|
| Guiding Language Translation with Translation Documents Using ML | US20260050753 | 2026 |
| Brand-Aligned Marketing Content Generation (Adobe Brand Intelligence) | US20250117829 | 2025 |
| Cultural Adaptation of Images Using Generative AI | US20260051084 | 2026 |
| Parameter-Efficient Fine-Tuned Models | US20250217193 | 2025 |
| Video-to-Video Culturalization Using Generative AI | Filed with USPTO | 2026 |
| Learning to Be Surprised: A Curiosity-Driven Framework for Creative Judgment | Filed with USPTO | 2026 |

## EXTERNAL CONTRIBUTIONS

**VibeNanoChat: GPT-2 Scale LLM Pretraining from Scratch** 2026 | code
Pretrained GPT-2-scale models from scratch at **>1M tokens/s** on 8× H100 using a custom distributed Muon optimizer. Derived compute-optimal scaling frontiers from isoFLOP curves over 40+ runs. Deep dives: Going Fast and Scaling Transformers.

**BrownBearAI: On-Device Photo Understanding for iOS** 2026 | App Store
Shipped a privacy-first iOS app executing on-device neural face clustering and deduplication at **99.4% BCubed F1** on LFW identities with zero cloud dependency.

**LaSeRo: Segmenting Antarctic Rock Outcrops from Landsat 8** 2019–2020 | code
Trained and evaluated SegNet architectures to segment geological rock outcrops across multispectral satellite imagery.

**Face Mask Detection for COVID-19** 2020 | code
Developed a real-time mask detection pipeline, benchmarking CNN architectures with VAE representation learning and Grad-CAM visual attribution.

## TECHNICAL SKILLS

**Research:** VLM Training & Evaluation, LLM Pretraining & Scaling Laws, Benchmark Design, Human Preference Studies, LLM-as-a-Judge & Reward Models, Reinforcement Learning (GRPO), Reward Design, Linear Probing, Bootstrap Statistics

**Tools:** Python, PyTorch, Distributed Training (DDP, ZeRO-2), torch.compile, FlashAttention-3, Liger Kernels, Muon, Mixed Precision (BF16), LoRA, Hugging Face Transformers/PEFT, Weights & Biases

## EDUCATION

| Institution | Degree | GPA | Years |
|---|---|---|---|
| Georgia Institute of Technology | M.S. Computer Science (Machine Learning) | 4.0/4.0 | 2019 – 2021 |
| BITS Pilani | B.E. Computer Science, M.Sc. Mathematics | 8.75/10 | 2011 – 2016 |